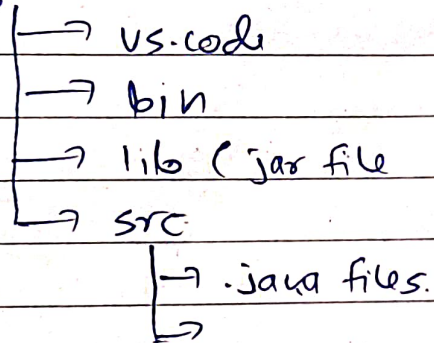


What is JDBC

- 1) JDBC is an java-based Application programming interface. that enables java application to connect with the Relational databases
- 2) It is used to retrieve and update data using SQL queries
- 3) It allows java application to perform CRUD operations by sending SQL commands to the database

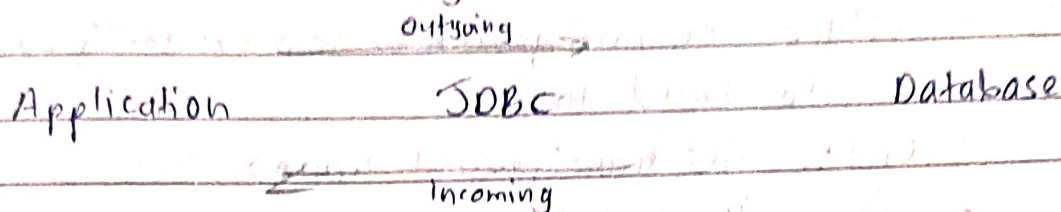
Folder structure.

Project Name



Java Database Connectivity (JDBC)

- it is API (Application programming Interface)
- it is used to connect java application with database (MySQL)



- it used to interact with database using CRUD operations (create, insert, update, Delete, select)

Stages OF JDBC (remove)

- 1) Load driver class (deprecated)
- 2) obtain a connection
- 3) Obtain a statements
- 4) execute the query
- 5) close the connection

how to connect Vscode + MySQL in JDBC

- 1) Vscode
- 2) Database (MySQL etc)
- 3) Driver (.jar)

How to create project in vs code.

setting → comand palette, → create java project →
No build tools → select folder → project Name.

~~folder structure~~ Project Name

Note → Username - root

Password - jishan@9136

imp

Agar java project section me project add nahi
hoga to wo project nahi hoga in uscode.

Page No.	
Date	

(Application)

How to create a project in JDBC (JDBC connector with Uscode)

- new folder
- open with uscode
- setting → Command palette → Java create java project
→ No build tools
- Enter file name (project Name)
- open with code (automatically)
- src - create java file (Java code)
- Now, in java projects → Referenced libraries →
go to plus (click) → C drive → program files x86 →
MySQL → connector → jar file.

uvimp Note - Jab tak apni log src me for file open nahi karke.

- jab tak java project folder file ka detect nahi karta.

in MySQL command line client

- write password that we have created at the time of
installing MySQL.

How to connect driver in eclipse with Jar file

- project that we have created → (right click) → properties →
Java build path → libraries → classpath → Add external jars → then
select location of jar file that we have downloaded → Apply →
Apply and close.

How to create the project -

File → New → project → java project.

// execute - execute karna (true or false return karayga)
if it gives (return) resultant true then it is
true other wise false

Simple JDBC create Database program

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.sql.Statement;
```

```
public class CreateDB {
```

```
    // throws ClassNotFoundException, SQLException
```

```
    String DB-URL = "jdbc:mysql://localhost/3306";
```

```
    String user = "root";
```

```
    String pass = "Jishan@9135";
```

// ① Load driver class (exceptional)

```
try {
```

// ② obtain a connection

```
Connection conn = DriverManager.getConnection(DB-URL,
user, pass);
```

// ③ obtain a statement

```
Statement stmt = conn.createStatement();
```

// ④ execute the query

```
String sql = "drop create database Demo";
```

```
stmt.executeUpdate(sql);
```

```
System.out.println("Database created successfully");
```

```
// 5) close the connection
conn.close();
```

```
3
```

```
catch (SQLException e) {
    e.printStackTrace();
}
```

```
3
```

Prepared statement interface

7 - prepared statement objects have some useful additional features

imp - instead of hard coding queries (such as values) it provides a feature to execute a parameterized query.

8 insert into stud

values (1, 10) → values are hardcoded in this query.
(2, 10)

better way - prepared statement

Syntax -

String sql = "insert into stud values (?, ?, ?)";

Ex=

String sql = "insert into stud values (?, ?, ?)";

PreparedStatement pstmt = conn.prepareStatement(sql);

pstmt.setInt(1, 10);

pstmt.setString(2, "shivam");

pstmt.executeUpdate();

System.out.println();

Note - Index is started from '1'

Methods -

1) setInt(param index, int value) → set the integer value to given parameter index.

2) setString("-", string value) → "-" string value -

3) setFloat("-", float value) → "-" float value

4) setDouble()

5)

↳ Kitni rows affected hui hai ye batata hai (in int format)

Note -

1) execute - it is used for all (update, insert, create)

2) executeUpdate → it is used for all (-1 -)

3) executeQuery → it is used for only iterate through to query (ex. selecting the and read the data from the table)

ResultSet interface

- it is an interface used to iterate through the data row by row.

- used when select query is used

- used to read the data line by line using get Method

ex=

ResultSet rs = stmt.executeQuery(sql);

while(rs.next())?

Here,

System.out.println(rs.getInt(1));

1, 2, 3 are the

System.out.println(rs.getString(2));

index of data

System.out.println(rs.getInt(3));

in the table

3.

Note -

When you are trying to update the table data using prepared statement we have to set data in order they are declared in the query.

To order the apun Ne query elika hai wohi order Me data changed karna hai.

Batch processing

- Instead of executing a single query, we can execute a batch (group) of queries.
- It makes the performance fast.

Syntax = (using statement interface)

```
conn.setAutoCommit(false);
statement st = conn.createStatement();
st.addBatch("insert into user values(190, 'abhi', 9000);");
st.addBatch(" -- ");
```

```
st.executeBatch(); // executing the batch
```

Syntax - (using prepared statement)

We have to take input from the user

Transaction Handling

1) conn.setAutoCommit(false) → automatic commit ko band karta hai (automatic call karna)

2) conn.commit() → apply all changes, saved all transaction and end transaction.

3) conn.rollback() → discard all changes and end transaction undo the transaction